Region-Wise Descriptive Analysis of MGNREGA in India (2018–2025): A Seven-Year Comparative Study on Employment Generation and Socio-Economic Trends

1. Introduction

The Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) is one of the largest public employment programs globally. It guarantees 100 days of wage employment per rural household annually and serves as a major safety net for rural livelihoods in India. The scheme plays a vital role in poverty reduction, income stabilization, asset creation, women's empowerment, and social inclusion.

2. Problem Statement

Although MGNREGA operates under a uniform national framework, significant variations exist in its implementation and outcomes across regions. Significant disparities exist across regions, states, and districts in terms of

- Employment generation
- Wage rates
- Labour budget allocation
- Expenditure efficiency
- Social inclusion
- Administrative efficiency

These differences arise due to variations in socio-economic conditions, governance capacity, agricultural dependency, migration patterns and regional development levels.

Over the years, particularly during the COVID-19 pandemic (2020–2021), MGNREGA played a critical role in absorbing reverse migration and providing emergency livelihood support. However, the magnitude and sustainability of this impact differed across regions.

There is a lack of structured, region-wise comparative studies that analyze long-term trends in employment generation and expenditure patterns over multiple years. Most existing analyses focus either on national aggregates or short-term policy reviews, leaving a research gap in longitudinal, region-based descriptive analysis.

Therefore, the central problem addressed in this study is:

How has MGNREGA performed across different regions of India from 2018 to 2025 in terms of employment generation, wage growth, budget allocation, and socioeconomic inclusion, and what regional disparities can be identified through systematic data analysis?

This study seeks to examine whether:

- Increased budget allocation translates into higher employment generation.
- Wage growth is consistent across regions.
- Social inclusion indicators (women participation) vary geographically.
- COVID-related employment spikes were uniform or region-specific.
- Certain regions show structural efficiency or inefficiency patterns.

3. Objectives of the Study

The primary objective of this project is to conduct a region-wise descriptive analysis of MGNREGA performance across six states (Andhra Pradesh, Rajasthan, Bihar, Assam, Maharashtra and Madhya Pradesh) representing six geographical regions of India from 2018 to 2025.

The study aims to evaluate employment generation, financial expenditure patterns, inclusiveness, and administrative efficiency using data-driven techniques.

Specific Objectives

1. Employment Generation Analysis

- To analyze the total number of households and individuals who worked under MGNREGA.
- To study person days generated (SC, ST, Women, Differently-abled).
- To evaluate whether the scheme is achieving its goal of 100 days of wage employment.
- To compare average days of employment provided per household across regions.

2. Financial Performance Evaluation

- To examine total expenditure trends over time.
- To analyze the split between Wage expenditure, Material and skilled wages, Administrative expenditure.
- To evaluate the ratio of wage-to-material expenditure.
- To study labour budget approval vs actual expenditure.

3. Inclusiveness & Social Equity Assessment

- To determine whether economically and socially vulnerable groups are adequately represented.
- To assess participation of Women, Scheduled Castes, Scheduled Tribes, Differentlyabled persons
- To compute participation percentages for fair comparison across states.

4. Administrative Efficiency Evaluation

- To analyze percentage of payments generated within 15 days.
- To assess fund utilization efficiency.
- To identify possible delays or inefficiencies in implementation.

5. Regional Comparison

- To compare performance indicators across six regions of India.
- To identify High-performing states , Low-performing states and Regional disparities
- To determine structural or policy-based differences influencing outcomes.

6. Trend Analysis (2018–2025)

To examine year-wise trend in:
• Employment generation
• Wage rate
• Expenditure
• Budget utilization
• To study the impact of major events (e.g., pandemic period) on employment demand and spending.

7. Feature Engineering

To create derived performance indicators such as:

- Expenditure per household
- Wage share ratio
- SC/ST participation percentage
- Women participation ratio
- Employment fulfillment ratio
- To standardize data for meaningful cross-state comparison.

4. <u>Basis for Topic Selection</u>

The topic was selected due to its socio-economic relevance and analytical potential. MGNREGA directly influences rural income stability and poverty reduction. Understanding its regional performance contributes to evidence-based policy evaluation.

The availability of structured district-level data allows application of data science techniques such as preprocessing, aggregation, growth rate analysis, and comparative visualization.

Some of the considerations for topic selection are:

1. National Relevance

MGNREGA is one of the largest public employment programs globally and an important factor of rural welfare policy in India. Studying its functioning provides insights into rural economic resilience, poverty alleviation mechanisms, and government intervention effectiveness.

2. Socio-Economic Significance

Rural India continues to face unemployment, income stability and migration pressures. MGNREGA acts as a safety net for vulnerable populations. Understanding its performance helps evaluate:

- Livelihood security
- Inclusion of marginalized communities
- Gender participation in rural employment

Thus, the topic directly connects to socio-economic development and public welfare.

3. COVID-19 Context

During the COVID-19 pandemic, MGNREGA became a primary employment source for returning migrant workers. Analyzing the 2018–2025 period captures:

- Pre-COVID baseline trends
- COVID employment surge
- Post-COVID stabilization

This makes the study temporally relevant and analytically rich.

4. Data-Driven Analytical Scope

The availability of structured administrative data allows for:

- Time-series analysis
- Regional comparison
- Growth rate computation
- Inclusion metrics analysis
- Efficiency evaluation

The dataset's monthly district-level granularity makes it suitable for:

- Descriptive analytics
- Comparative statistical evaluation
- Feature engineering
- Visualization-driven storytelling

The topic was chosen due to its national relevance, socio-economic importance, data availability, and analytical depth. By conducting a structured region-wise descriptive analysis of MGNREGA from 2018 to 2025, the study aims to generate meaningful insights into employment generation patterns and policy effectiveness across diverse Indian regions.

5. Project Vision and Analytical Approach

Vision

The vision of this project is to transform raw administrative records into meaningful regional insights that highlight employment trends, financial allocation patterns, and socioeconomic inclusion under MGNREGA.

The study emphasizes descriptive analytics, trend analysis, and comparative evaluation rather than predictive modeling, focusing on understanding patterns and disparities over time.

Methodological Approach

The project follows a structured analytical framework:

- Data integration and validation
- Data preprocessing and cleaning
- Zero-value treatment and consistency checks
- Feature engineering
- Aggregation at state-year and region-year levels
- Exploratory Data Analysis (EDA)
- Trend interpretation and comparative assessment
- Insight generation and policy discussion

Tools Used

- Python
- Jupyter Notebook
- PowerBi
- Pandas
- Numpy
- Matplotlib & Seaborn
- Scipy

The study applies descriptive statistics, time-series comparison, growth rate computation, and index creation techniques in the later stages.

6. Dataset Details

Dataset Source

The dataset is derived from official administrative records of MGNREGA published by the

Government of India. It contains district-level monthly data for six selected states namely Andhra Pradesh, Assam, Bihar, Rajasthan, Maharashtra and Madhya Pradesh representing six regions of India based on the maximum participation in the scheme.

Source Link: https://www.data.gov.in/resource/district-wise-mgnrega-data-glance

Nature of Data

- Structured tabular dataset
- Monthly time-series cross-sectional data
- Quantitative administrative records
- Multi-dimensional indicators related to employment and expenditure

Dataset Size

- Total Rows: 72,575
- Total Columns: 36
- Time Period: 2018–2025
- Coverage: Six states across six regions (North, South, West, East, North-East, Central India)
- Memory Usage: Approximately 20 MB

7. Description of Important Variables

1. Total_Exp
Represents the total expenditure incurred under MGNREGA in a given state and year. It reflects overall financial utilization of allocated funds.

2. Wages
Total amount spent on wage payments to workers. This indicates the direct income support provided to rural households.

3. Material_and_Skilled_Wages
Expenditure on materials and payments to skilled labor. It helps assess compliance with the wage-to-material ratio rule under MGNREGA.

4. Total_Adm_Expenditure
Administrative expenses incurred in implementing the scheme. High values may indicate higher operational costs.

5. Approved_Labour_Budget
The officially approved labour budget for a state/year. It represents planned employment generation targets.

6. Total_Individuals_Worked

Total number of individuals who worked under the scheme. It shows overall participation.

7. Total_Households_Worked
Number of households that received employment. Important for measuring rural coverage.

8. Total_No_of_HHs_completed_100_Days_of_Wage_Employment
Number of households that completed the full 100 days of guaranteed employment. Indicates success in fulfilling the scheme's core objective.

9. Average_days_of_employment_provided_per_Household
Average number of days of employment provided per household. Reflects employment intensity.

10. Women_Persondays
Total persondays generated by women workers. Used to assess gender participation and empowerment.

11. SC_persondays
Persondays generated by Scheduled Caste beneficiaries. Measures social inclusion.

12. ST_persondays
Persondays generated by Scheduled Tribe beneficiaries. Important for evaluating support to tribal populations.

13. Differently_abled_persons_worked
Number of differently-abled individuals who participated. Reflects inclusiveness of the scheme.

14. Persondays_of_Central_Liability_so_far
Total persondays funded under central government liability. Indicates central financial responsibility.

15. Average_Wage_rate_per_day_per_person
Average daily wage paid per worker. Useful for income analysis and wage trend study.

16. Number_of_Completed_Works
Total number of works completed under MGNREGA. Indicates asset creation performance.

17. percent_of_NRM_Expenditure
Percentage of expenditure on Natural Resource Management works. Reflects focus on sustainable rural development.

18. percent_of_Expenditure_on_Agriculture_Allied_Works
Percentage of funds spent on agriculture-related works. Shows contribution toward strengthening rural livelihoods.

19. percentage_payments_generated_within_15_days
Percentage of wage payments processed within 15 days. Indicates administrative efficiency and compliance.

20. Number_of_GPs_with_NIL_exp
Number of Gram Panchayats with zero expenditure. May indicate implementation gaps or inactivity in certain regions. These variables allow evaluation of employment intensity, inclusion, expenditure efficiency, and wage trends.

Target Variable

This study is descriptive in nature and does not involve predictive modeling. Therefore, no specific target variable is defined. However, primary outcome indicators include:

- Average wage rate per day per person
- Total Expenditure
- Approved Labour Budget
- Total Individuals Worked

8. Data Preprocessing

1. Null Value Analysis

The dataset was verified for missing values and none were found.

2. Zero Value Treatment

Several columns contained zero values. These were categorized as:

- Valid structural zeros (e.g., no SC participation in a given month).
- Financial zeros requiring verification.
- Zero values were first replaced with NaN (missing values).
- The missing values were then imputed using the median value within each state and financial year group.
- This approach was adopted because the dataset contains extreme outliers and median provides a more robust and representative measure of central tendency for imputation.
- This preserved regional characteristics while correcting inconsistencies.

The treatment was applied to the following variables:

- Approved_Labour_Budget
- Average_Wage_rate_per_day_per_person

3. Logical Consistency Check

One record showed 771 individuals worked while the average wage rate was recorded as zero, which is logically inconsistent because workers cannot work without wages. Therefore, this record was removed from the dataset. Remaining zero wage values correspond to cases where no individuals worked and are retained as valid observations.

4. Data Type Standardization

To improve analytical efficiency and ensure correct statistical operations, the data types of several variables were standardized. Categorical conversion was applied to nominal variables. Ordered categorical format was applied to the month variable to maintain the correct financial year sequence.

- Financial year converted to categorical format.
- Month converted to ordered categorical (April–March).
- State and district converted to categorical types.

This conversion reduced memory usage and ensured correct ordering for times-series analysis.

5. Duplicate Validation

Records were verified to ensure uniqueness based on (State, District, Financial Year, Month) and 4 duplicate entries were detected and removed using drop_duplicates ( ). Data integrity was preserved.

6. Column Name Standardization

During inspection, one variable name contained a spelling inconsistency which could cause errors during analysis.

The column name 'percentage_payments_gererated_within_15_days' was corrected using a renaming operation to 'percentage_payments_generated_within_15_days'.

7. Removal of Irrelevant Columns

Columns that were not relevant for analytical purposes were removed to simplify the dataset structure. Remarks, state_code, district_code and month columns were dropped which reduced dataset dimensionality.

9. Advanced Preprocessing

1. Logical Range Validation

Logical validation of percentage-based variables revealed that the variable percentage of payments generated within 15 days contained values exceeding 100%, which is not feasible. These values were treated as data inconsistencies and replaced using median imputation to maintain the integrity of the dataset.

2. Outlier Detection

During exploratory analysis, extremely high values were observed in the variable Average Wage Rate per Day per Person. The maximum value exceeded ₹8.8 million, which is unrealistic in practical scenarios.

Instead of applying global outlier detection, a group-wise Interquartile Range (IQR) method was used within each state. This approach preserves regional variations in labour budgets and wage structures while removing extreme anomalies caused by reporting errors or denominator distortions.

Capping approach was applied on Approved Labour Budget to reduce the influence of extreme observations while preserving the dataset. Capping is a method where extreme values are limited to a threshold instead of being removed helping preserve data while reducing distortion.

After applying the state-wise IQR method, extreme anomalous values were successfully removed, reducing the dataset to 63,028 rows. The maximum average wage rate is now ₹370, which falls within a realistic range, compared to the earlier unrealistic value of ₹88 lakh.

This confirms that the outlier treatment has improved data reliability while preserving meaningful variations.

3. Creation of Time Variable

A numeric year variable was extracted from the financial year column, ‘fin_year’ to enable easier trend analysis and plotting.

4. State-Year Aggregation

The dataset originally contained district-level monthly data.
To analyze broader patterns, the data was aggregated to the state-year level.
Aggregation functions used:

- Sum for cumulative indicators
- Mean for averaged indicators

The variables used are:

- Total_Exp
- Approved_Labour_Budget
- Average_Wage_rate_per_day_per_person
- Total_Households_Worked
- Total_Individuals_Worked
- Average_days_of_employment_provided_per_Household
- Persondays_of_Central_Liability_so_far
- SC_persondays
- ST_persondays
- Women_Persondays
- Differently_abled_persons_worked
- Wages
- Material_and_skilled_Wages
- Total_Adm_Expenditure
- Total_No_of_HHs_completed_100_Days_of_Wage_Employment
- Total_No_of_JobCards_issued
- Number_of_Completed_Works

- Total_No_of_Works_Takenup
- percentage_payments_generated_within_15_days

Therefore, a state-year dataset was created which reduced dataset complexity while preserving key indicators and enabling cross-state comparison.

5. Region-Year Level Aggregation

To perform regional comparative analysis, state-level data was further aggregated into region-year level data.
Each state was mapped to its respective geographical region:

- North - Rajasthan
- South - Andhra Pradesh
- West - Maharashtra
- East - Bihar
- NorthEast - Assam
- Central - Madhya Pradesh

The same aggregation functions (sum and mean) used in state-level aggregation were applied which created a Region-Year dataset enabling region-wise comparative analysis across the study period.

This dataset contains Financial, employment, inclusiveness, administrative efficiency metrics and operational performance metrics.

10. Feature Engineering

1. Expenditure per Household

A new variable was created to measure financial expenditure relative to the number of participating households using Total_Exp and Total_Households_Worked.

- Indicates average financial spending per beneficiary household.
- Helps assess expenditure intensity across regions.

2. Wage Share Ratio

The proportion of total expenditure spent on wages was calculated using Wages and Total_Exp.

- Evaluates compliance with the wage-material ratio rule under MGNREGA.
- Higher values indicate stronger direct income support.

3. SC/ST Participation Ratio

Measures the share of employment generated for Scheduled Castes and Tribes, indicating the inclusiveness of MGNREGA, calculated using Persondays of SC/ST workers and Total_Persondays.

- Enables fair comparison across states by using proportional representation instead of absolute values.
- Helps assess effectiveness of policy outreach and identify gaps in inclusion.

4. Women Participation Ratio

The share of total persondays contributed by women workers was calculated using Women_Persondays and Persondays_of_Central_Liability_so_far.

- Measures gender inclusion in rural employment.
- Enables comparison of women's participation across regions.

5. Differently-Abled Participation Ratio

A participation ratio for differently-abled workers was created using Differently_abled_persons_worked and Total_Individuals_Worked.

- Evaluates inclusiveness of the program.
- Highlights participation of marginalized groups.

6. Employment Fulfillment Ratio

MGNREGA guarantees 100 days of employment per household.
A ratio was created to measure how much of this target was achieved using Average_days_of_employment_provided_per_Household.

- Indicates effectiveness in delivering the guaranteed employment target.

7. Labour Budget Utilization Ratio

A ratio was created to measure how effectively approved labour budgets were utilized using Total_Exp and Approved_Labour_Budget.

- Evaluates budget utilization efficiency.
- Helps identify under-utilization or overspending trends.

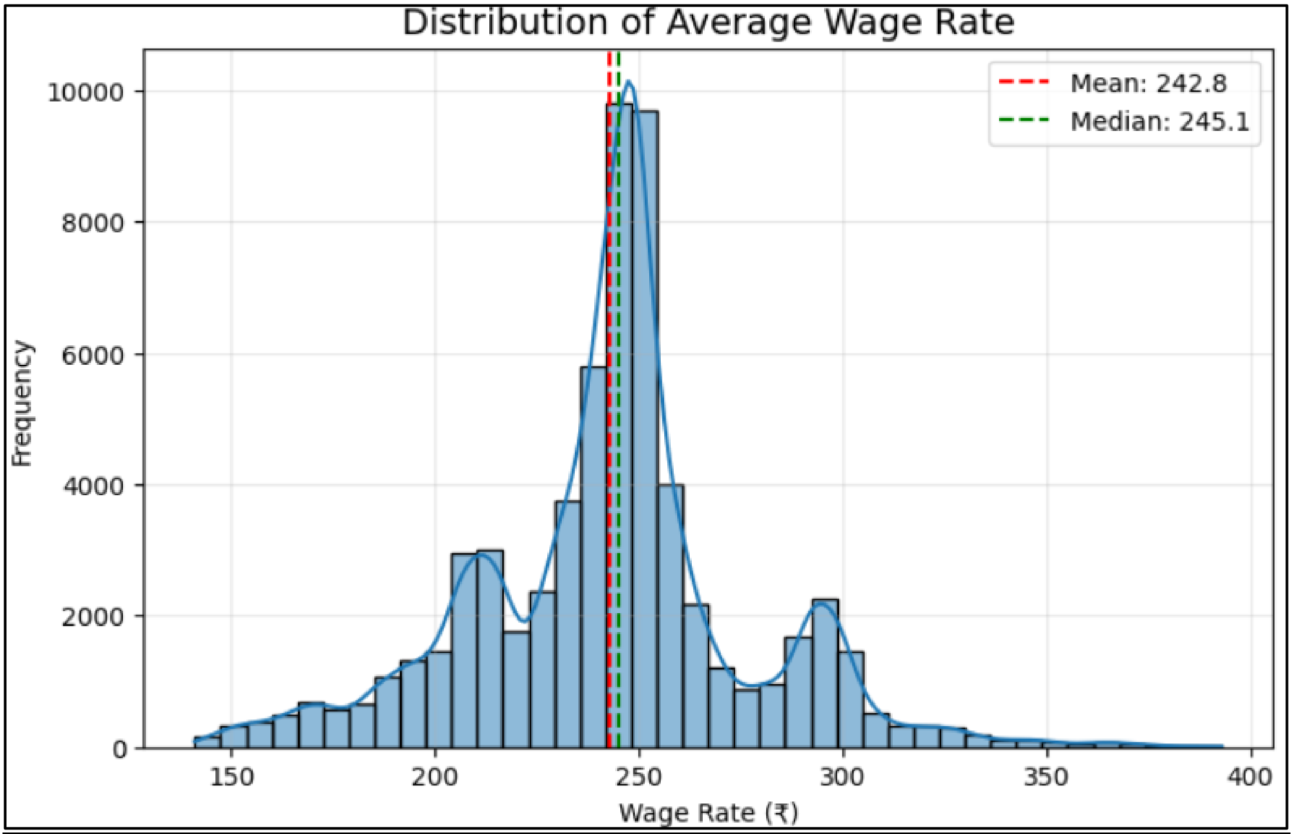
8. Work Completion Efficiency

Measures how effectively initiated works are completed under MGNREGA, calculated using Number_of_Completed_Works and Total_No_of_Works_Takenup.

- Indicates administrative and execution efficiency.
- Helps identify delays or inefficiencies in project implementation across states.

11. EDA and Visualizations

1. Wage Distribution Plot

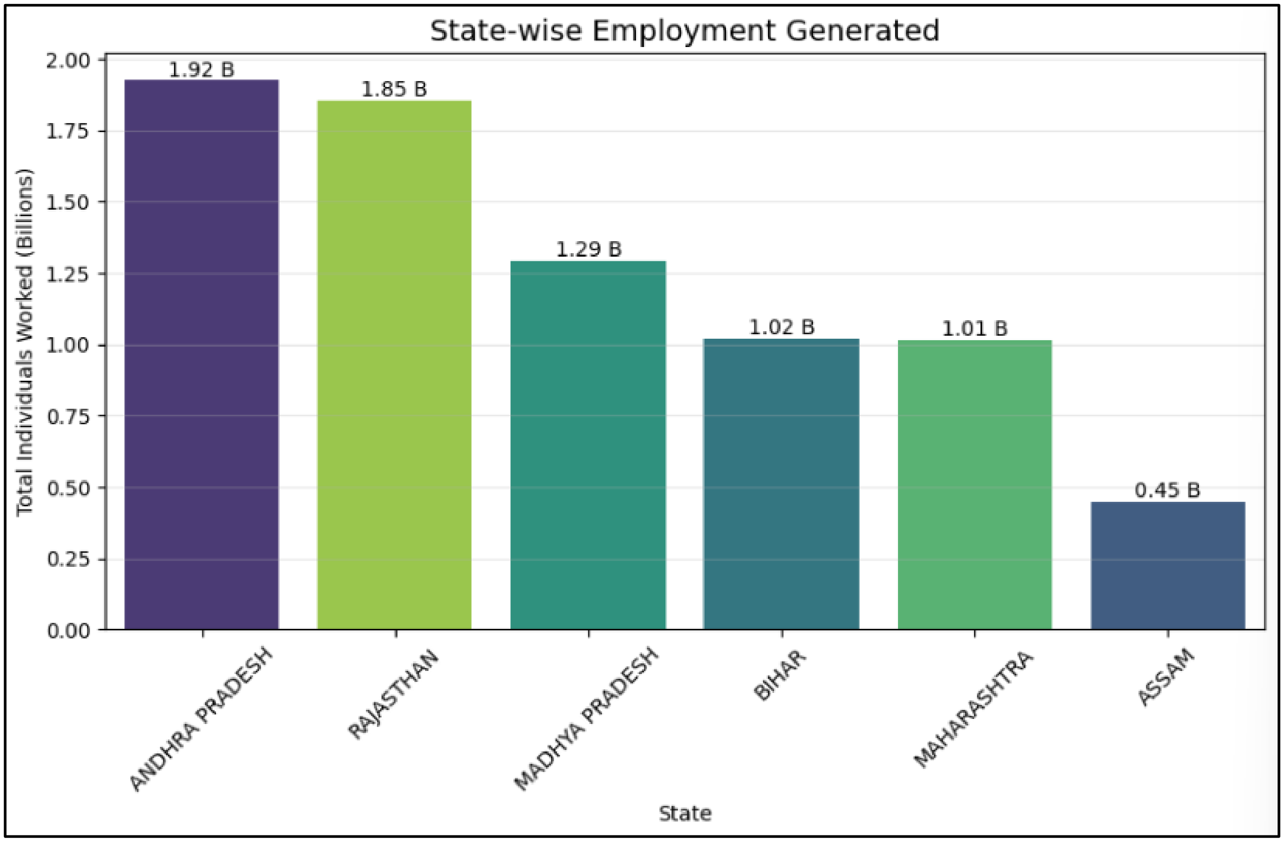


The distribution of average wage rate per day shows a concentrated range of values, indicating successful removal of extreme anomalies during preprocessing. Wage distribution is concentrated and consistent, with minimal skewness, indicating effective data cleaning and standardized wage patterns.

Insight:

- Mean wage ≈ ₹242.8 and Median ≈ ₹245.1 → values are very close, indicating a balanced distribution.
- Majority of wages are concentrated around ₹230 – ₹260, showing high consistency in wage rates.

2. State-Wise Employment Barplot

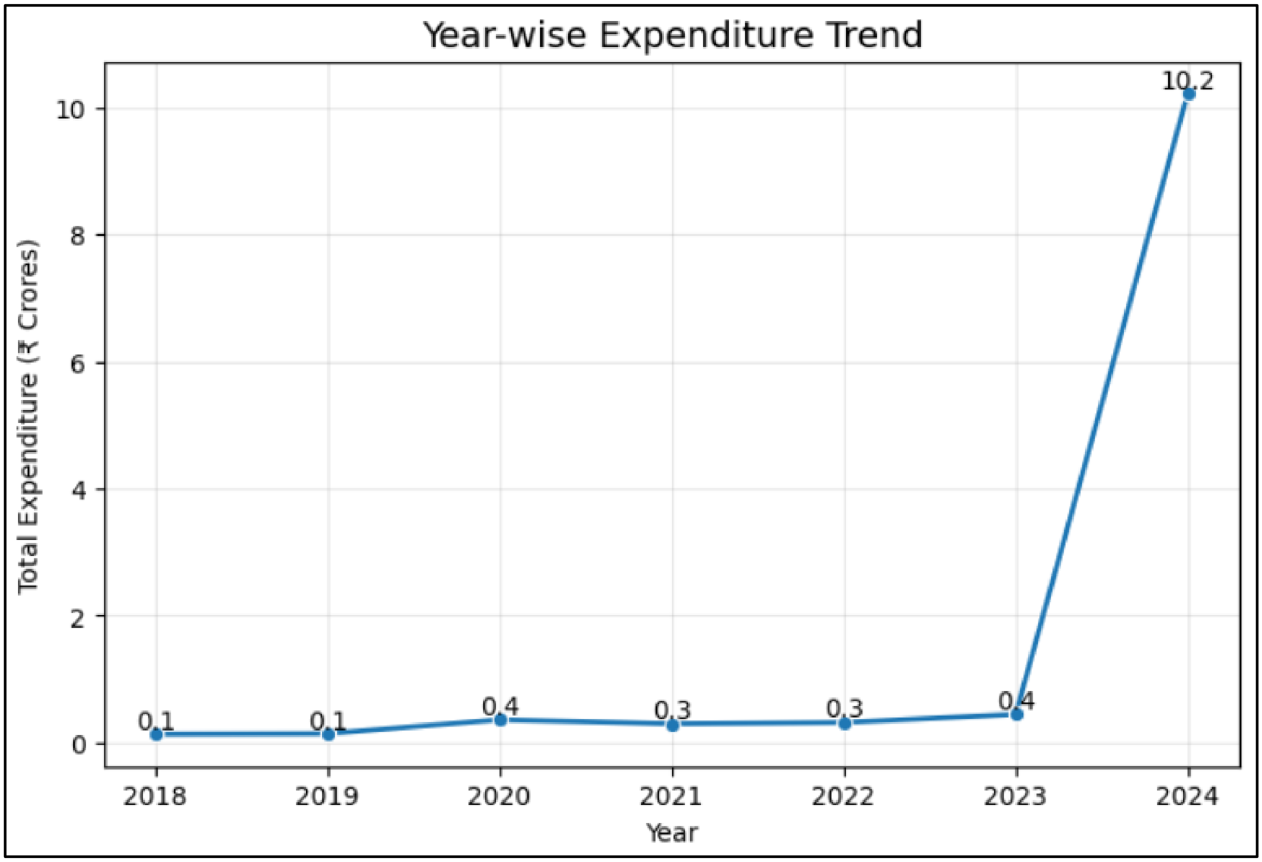


There is a significant variation in employment generation across states, indicating differences in demand for work and implementation efficiency. Some states contribute disproportionately to total employment under the scheme. States with higher employment figures indicate greater dependence on MGNREGA for livelihood generation.

Insight:

- Andhra Pradesh (1.92 B) and Rajasthan (1.85 B) generate the highest employment so they are the leading contributors.
- Madhya Pradesh (1.29 B) shows moderate employment, significantly lower than top states.
- Bihar (1.02 B) and Maharashtra (1.01 B) have almost similar employment levels.
- Assam (0.45 B) generates the lowest employment which shows comparatively low participation.
- There is a wide gap (~1.5 B) between highest and lowest states showing strong regional disparity.

3. Year-Wise Trend Plot

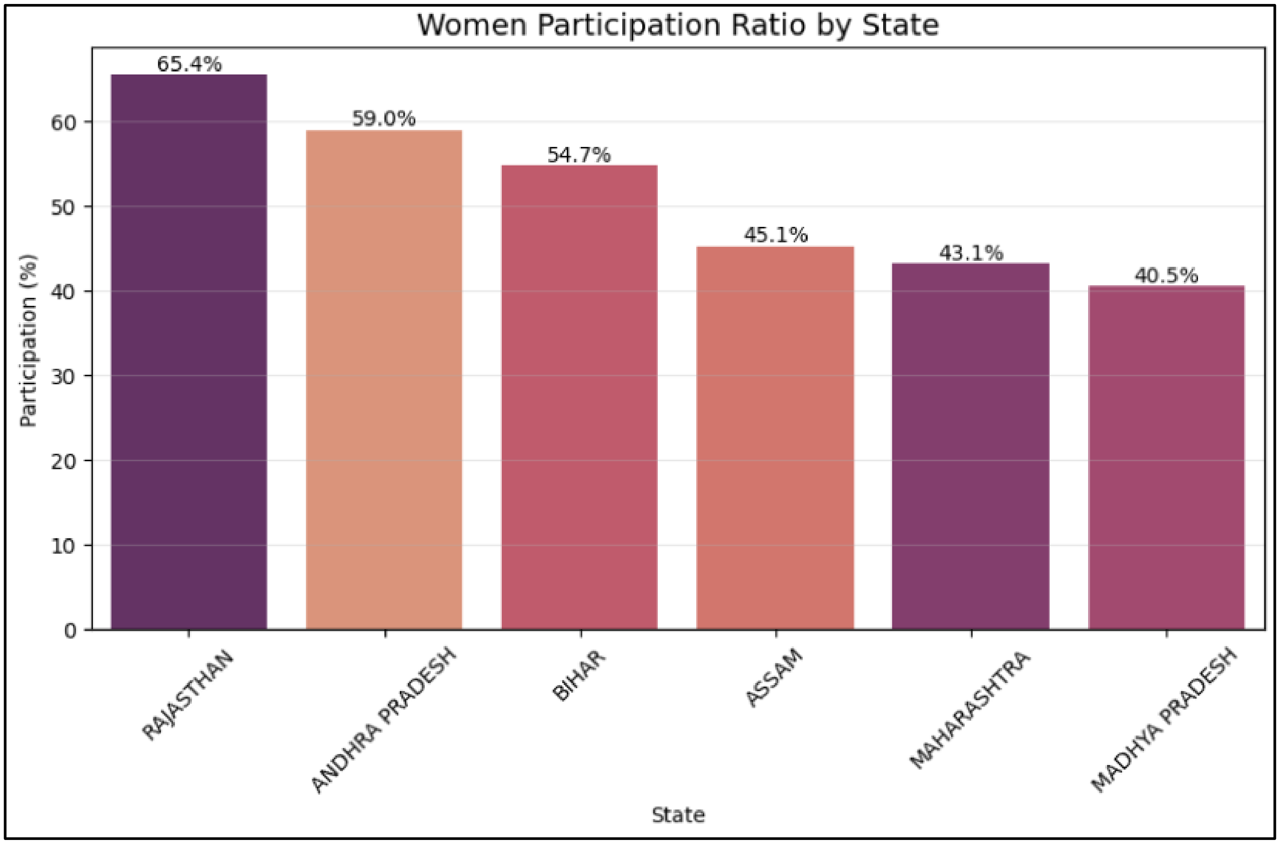


The trend shows fluctuations in total expenditure over the years, reflecting changing economic conditions and varying demand for employment. This indicates that the scheme responds dynamically to socio-economic needs or a sudden increase in financial allocation or reporting in the latest year.

Insight:

- Expenditure jumps drastically in 2024 indicating possible data imbalance or surge in funding
- Stable trend earlier shows consistent scheme implementation

4. Women Participation Ratio Bar Plot

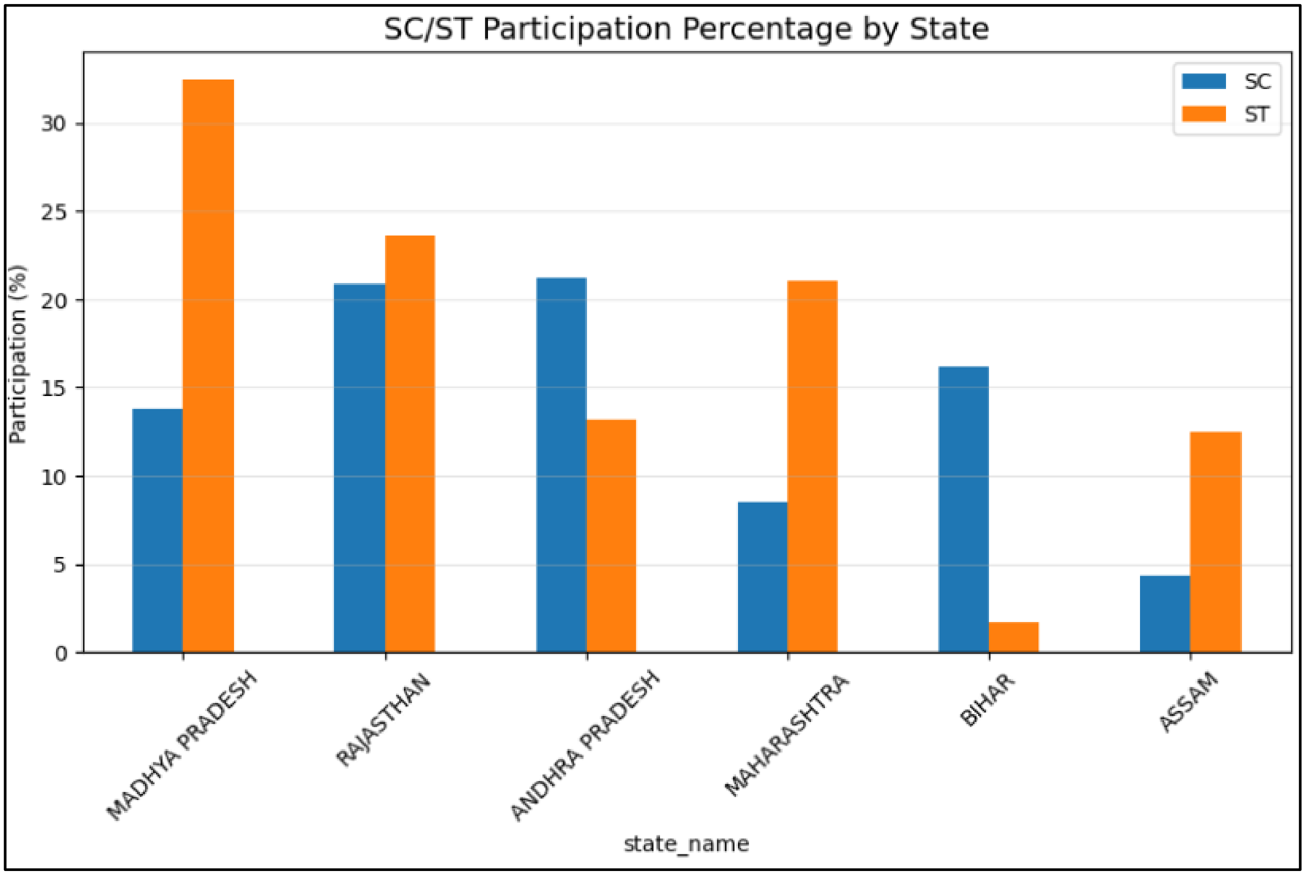


Women participation differs significantly across states, indicating varying levels of gender inclusivity. Higher participation reflects better alignment with the scheme's objective of empowering women.

Insight:

- Rajasthan (~65%) leads with strong gender inclusivity
- Madhya Pradesh (~40%) has lowest participation which shows the scope for improvement in women workforce.
- Reflects policy effectiveness in empowering women

5. SC/ST Participation % Bar Plot

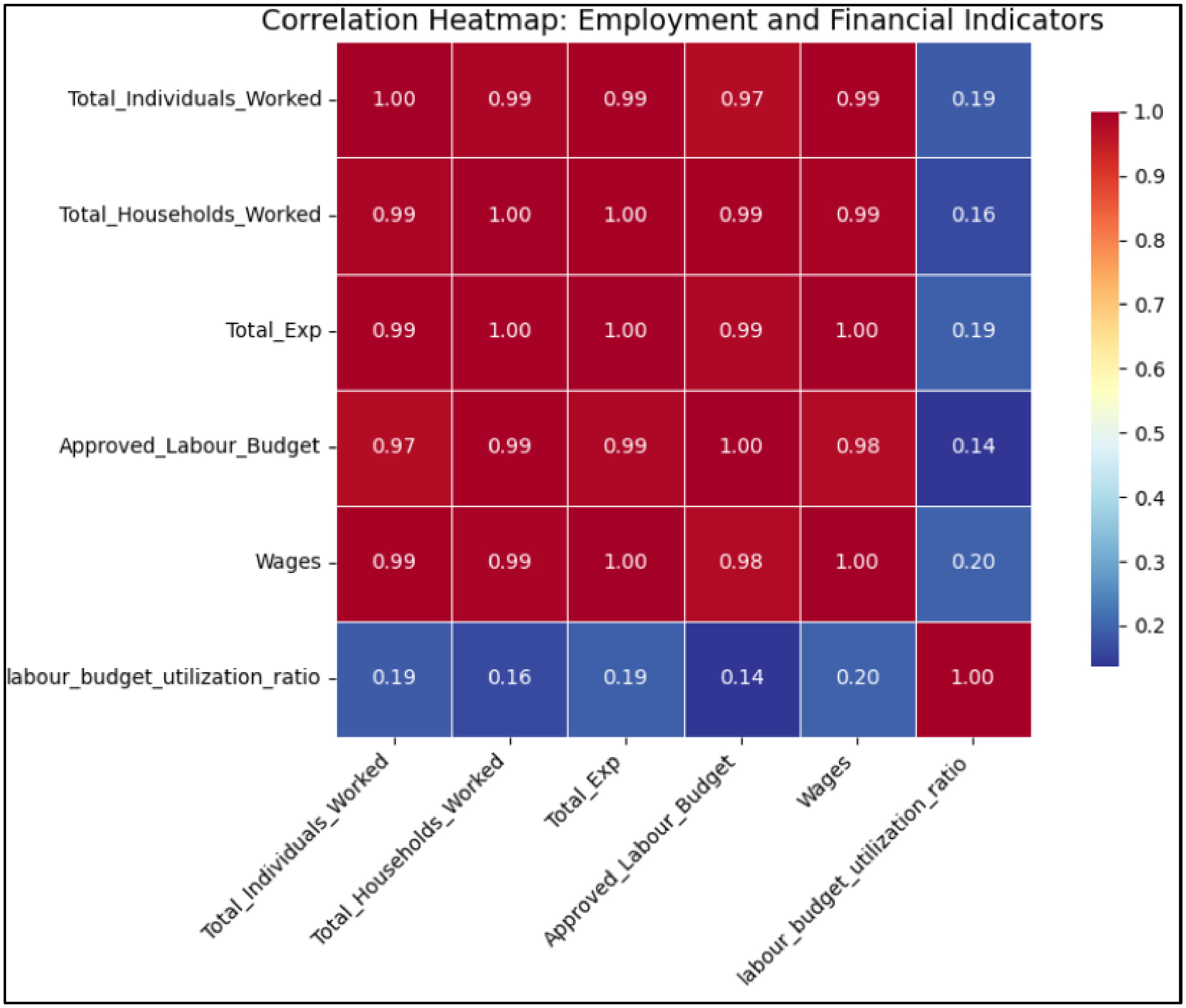


The participation of SC and ST communities shows noticeable variation across states, highlighting disparities in social inclusion. This suggests the need for targeted interventions in regions with lower participation.

Insight:

- MP has highest ST participation (~32%) indicating strong tribal inclusion
- AP shows higher SC participation (~21%)
- Bihar has very low ST showing regional inequality

6. Correlation analysis of Employment and Financial Indicators

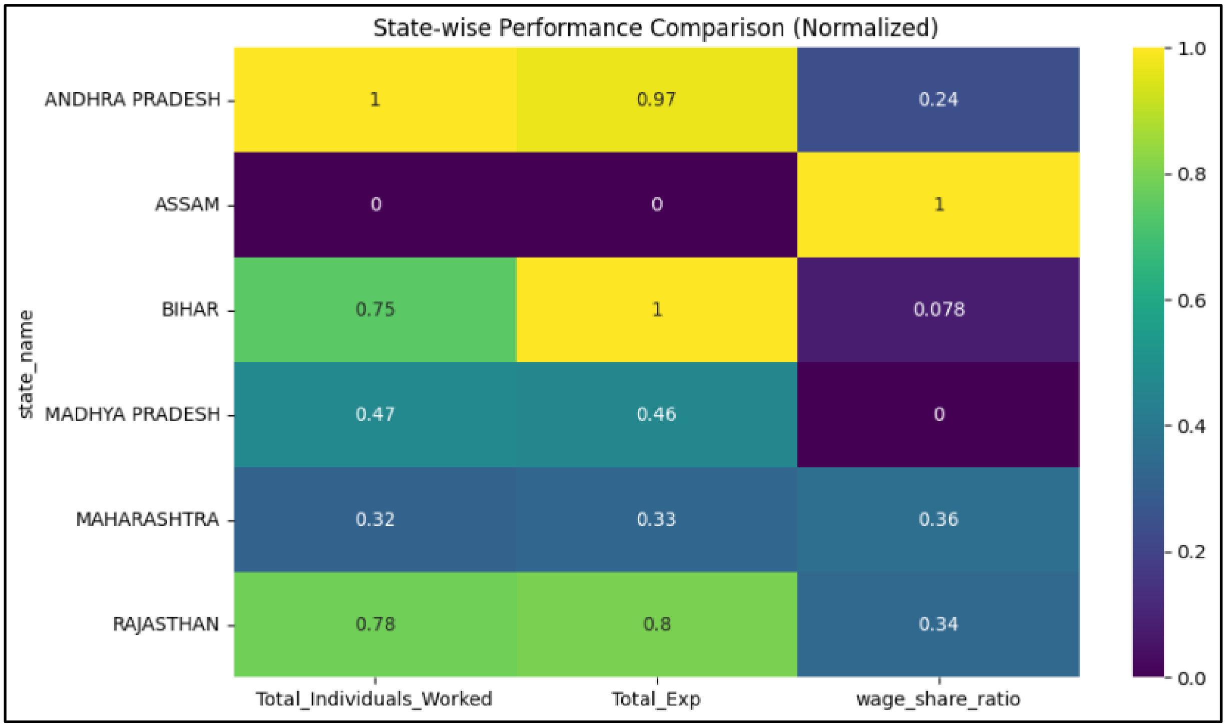


Strong positive correlations between expenditure, employment, and persondays indicate that higher financial allocation leads to increased employment generation. This highlights the interconnected nature of scheme performance indicators.

Insight:

- Very strong positive correlation (~0.97–1.00) between Total Expenditure, Wages, Households Worked, and Individuals Worked indicates that higher spending directly drives higher employment generation.
- Wages and Total Expenditure (~1.00) are almost perfectly correlated confirms that major portion of expenditure is allocated to wages, aligning with MGNREGA's objective.
- Approved Labour Budget (~0.97–0.99) is strongly correlated with actual expenditure and employment
- Labour Budget Utilization Ratio shows weak correlation (~0.14–0.20) with all major variables implies that efficiency of utilization is independent of scale of spending or employment.

7. Multi-Indicator State Comparison Heatmap

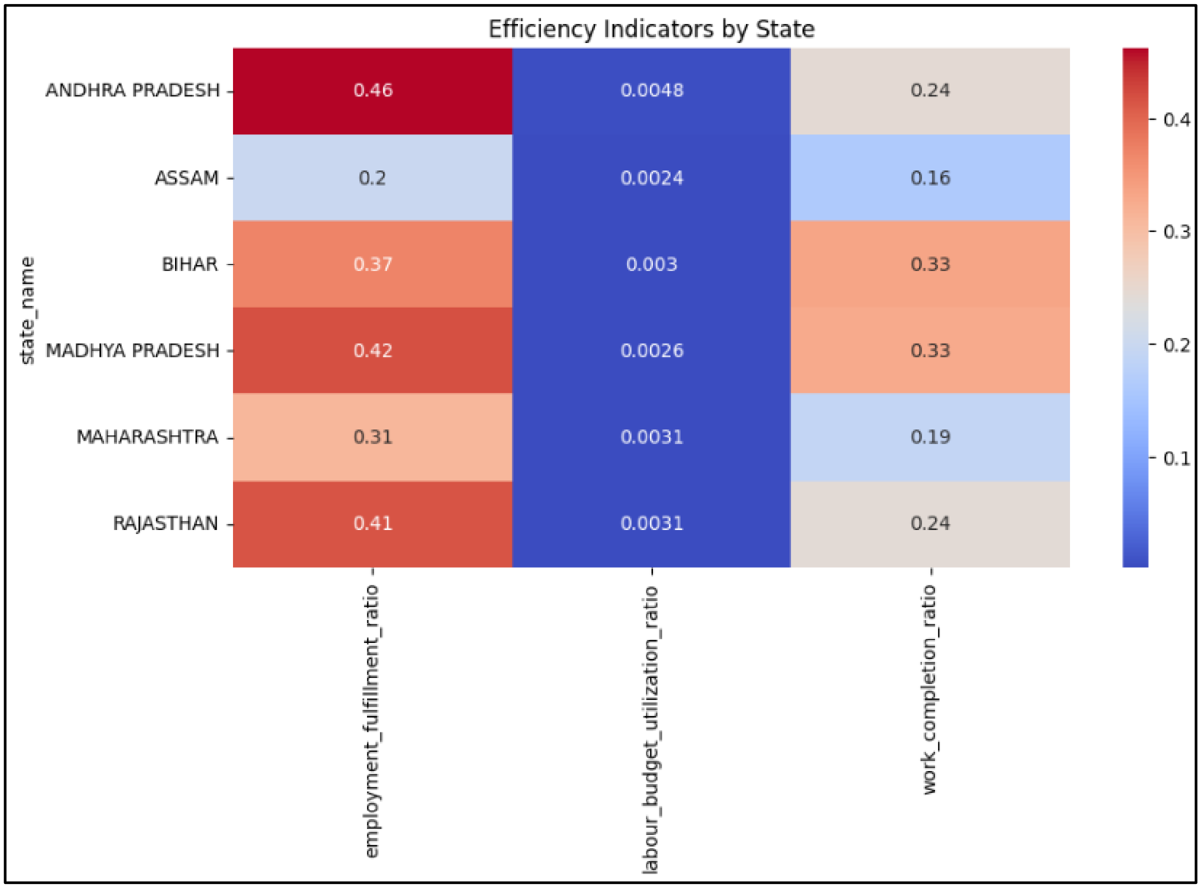


States differ in performance across employment, expenditure, and wage share, showing uneven development. While states like Andhra Pradesh and Rajasthan achieve both high scale and balanced efficiency, others like Assam and Bihar show skewed performance, indicating differences in implementation strategy and resource utilization.

Insight:

- Andhra Pradesh and Rajasthan emerge as top-performing states, showing high employment and expenditure with balanced wage allocation. This indicates effective conversion of financial resources into employment generation.
- Bihar shows high expenditure but low wage share, suggesting more spending on materials or administrative components rather than direct employment. This reflects an imbalance in resource allocation.
- Assam operates at a very low scale but with the highest wage share, indicating limited project scope with higher dependence on wage payments. This highlights inefficiency in scaling the scheme.
- Overall, there is a clear variation in performance across states, with differences in both scale and efficiency.

8. Efficiency Heatmap across States



Efficiency metrics vary across states, with Andhra Pradesh and Rajasthan performing better in employment fulfillment. Budget utilization remains low across all states.

Insights:

- Andhra Pradesh shows the highest employment fulfillment (~0.46), indicating strong and efficient implementation.
- Labour budget utilization is extremely low (~0.002–0.004) across states, highlighting inefficiency in fund usage.
- Work completion ratios are moderate, suggesting delays or gaps in execution of projects.
- Overall, efficiency varies across states, with scope for improving utilization and completion rates.

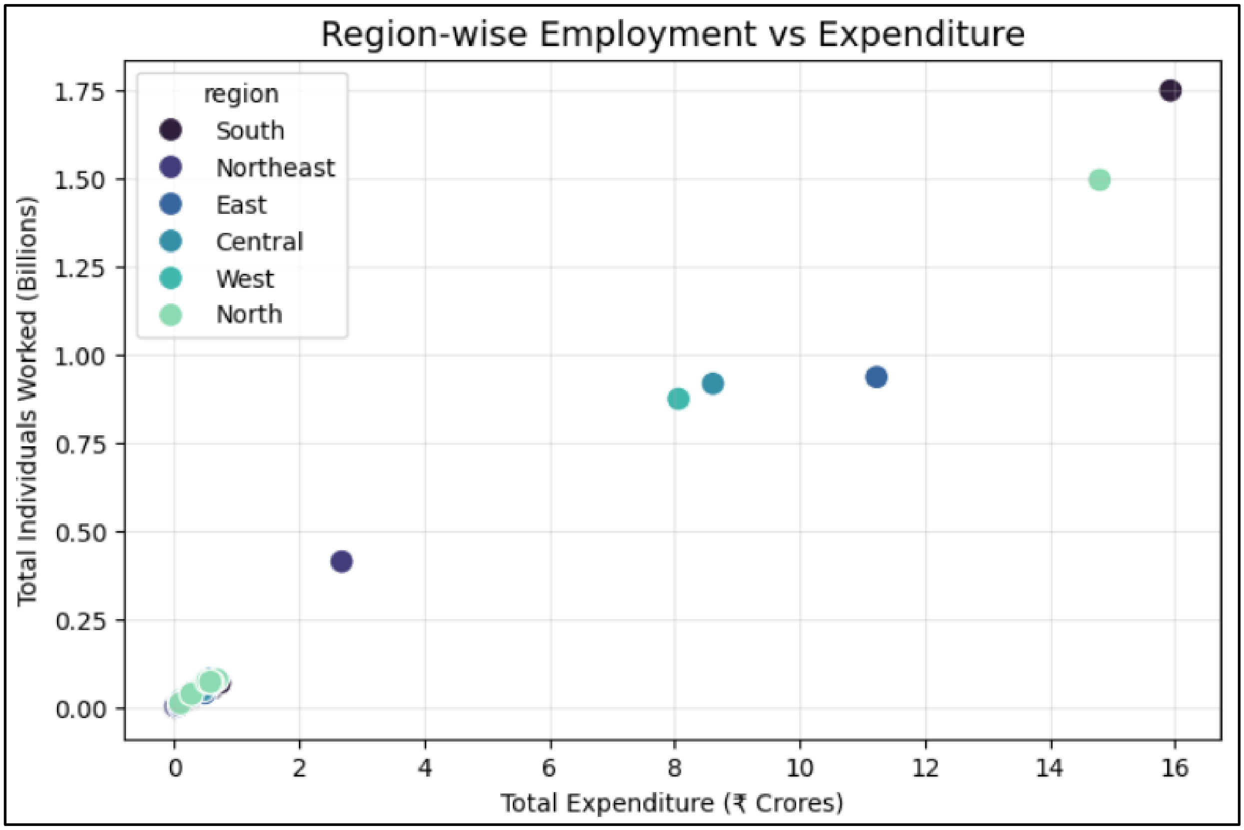
9. Regional Performance of Multiple Indicators Heatmap

Regions show clear disparities in performance, with South and East regions leading across most indicators. Northeast lags significantly across all metrics.

Insights:

- South region is the strongest performer, with high expenditure, employment, and participation.
  This reflects efficient and balanced implementation.
- Northeast region is the weakest, showing low scale and participation.
  Indicates need for policy focus and better execution.
- North region leads in women participation, highlighting strong gender inclusion.
  Shows effectiveness in social outreach.
- Overall, clear regional disparities exist in scale and inclusiveness.

10. Region-Wise Employment vs Expenditure Scatter Plot

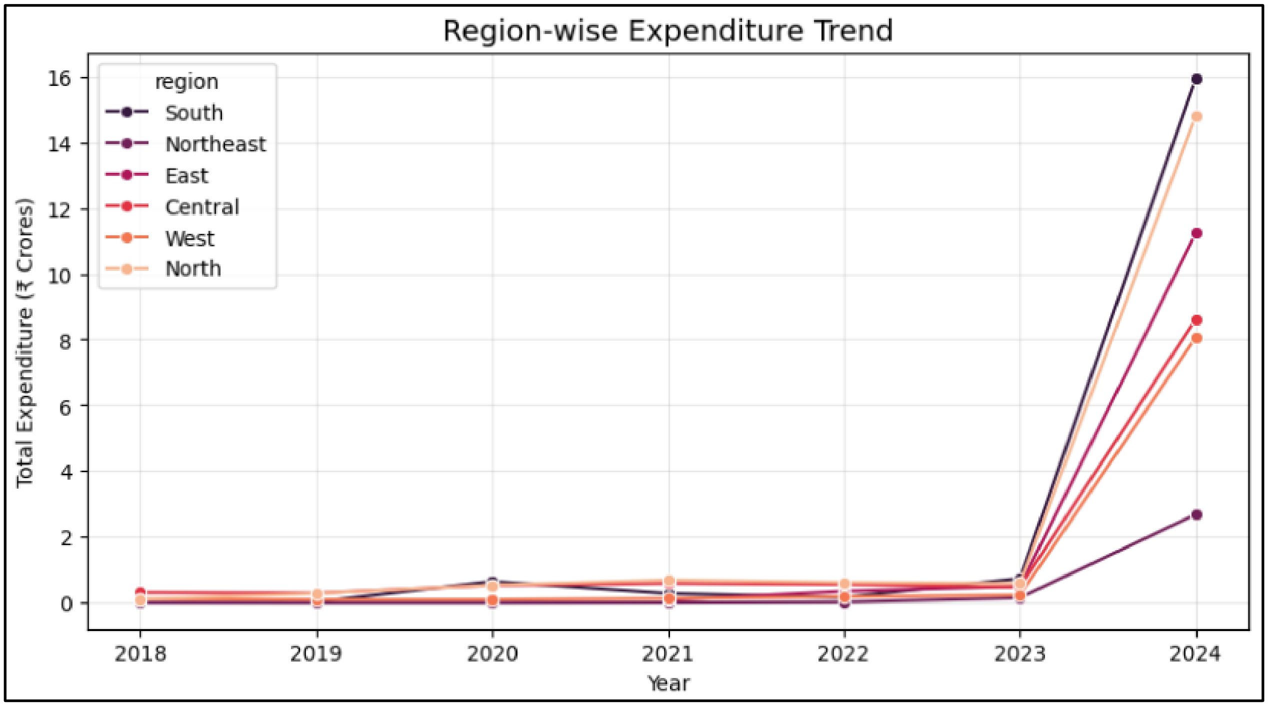


There is a strong positive relationship between expenditure and employment across regions, indicating that higher spending leads to higher job generation. However, efficiency varies as some regions generate more employment per unit of expenditure.

Insights:

- South and North regions show the highest expenditure and employment levels, making them dominant performers.
  This indicates strong implementation capacity and effective conversion of funds into employment.
- Northeast region records the lowest expenditure and employment levels, significantly lagging behind other regions.
  This reflects limited scale of operations and weaker scheme penetration.
- Central and West regions exhibit moderate expenditure and employment, indicating balanced but not leading performance.
  These regions show steady implementation but still fall short of top-performing regions.
- Overall, there exists a clear regional disparity in both scale and performance of MGNREGA.

11. Regional Trend Analysis

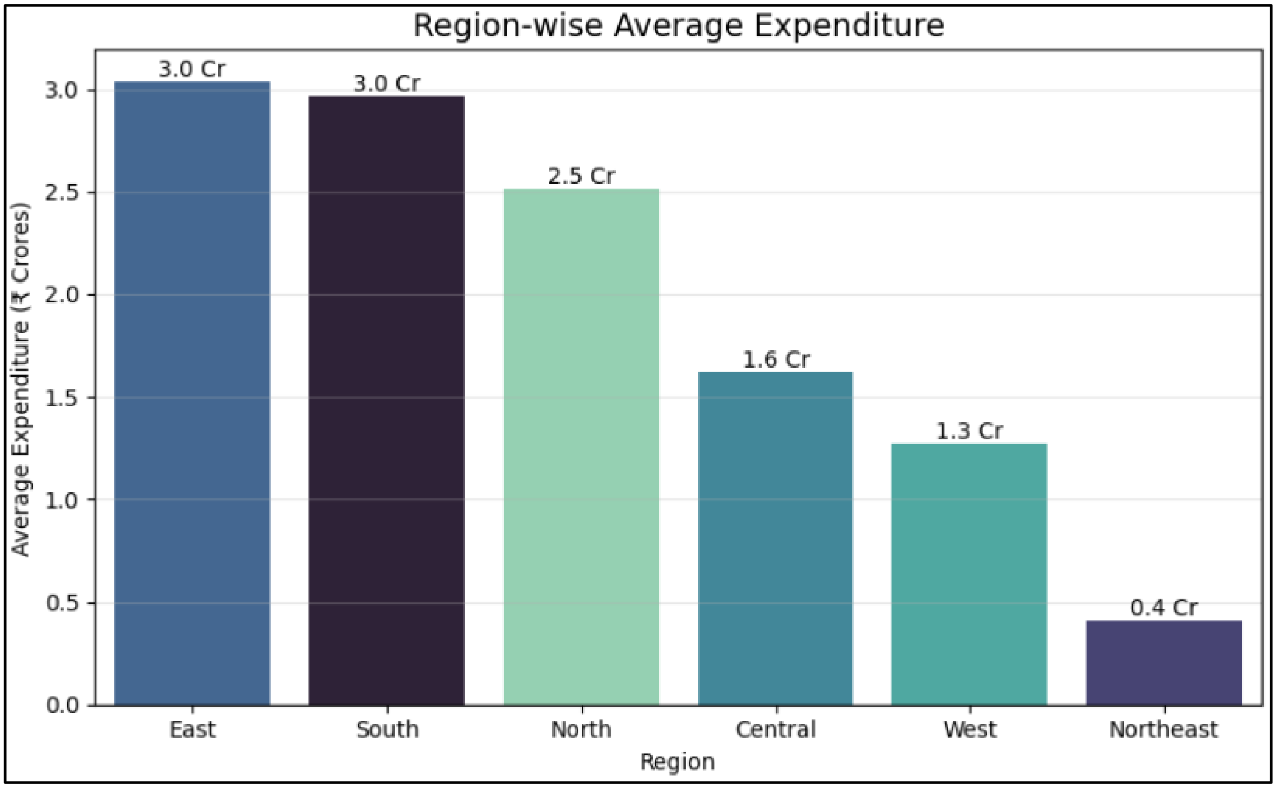


All regions show relatively stable expenditure until 2023, followed by a sharp surge in 2024. This indicates a significant increase in funding or reporting across all regions.

Insights:

- South and North regions dominate in 2024 expenditure, reaching the highest values (~₹16Cr and ~₹14–15Cr respectively).
  This indicates stronger implementation capacity and higher scale of MGNREGA operations in these regions.
- Northeast remains the lowest performing region, with expenditure around ~₹2–3 Cr even after the 2024 increase.
  This highlights a persistent regional gap and potential issues in fund allocation or execution.
- All regions show a sudden spike in expenditure in 2024, compared to relatively stable trends from 2018–2023 (~₹0–1Cr range).
  This suggests a possible policy push, increased funding, or data imbalance in the latest year.
- Central and West regions show moderate growth, reaching around ~₹8–9 Cr in 2024. This indicates gradual improvement but still lagging behind leading regions like South and North.

12. Region-Wise Expenditure Ranking Bar Plot

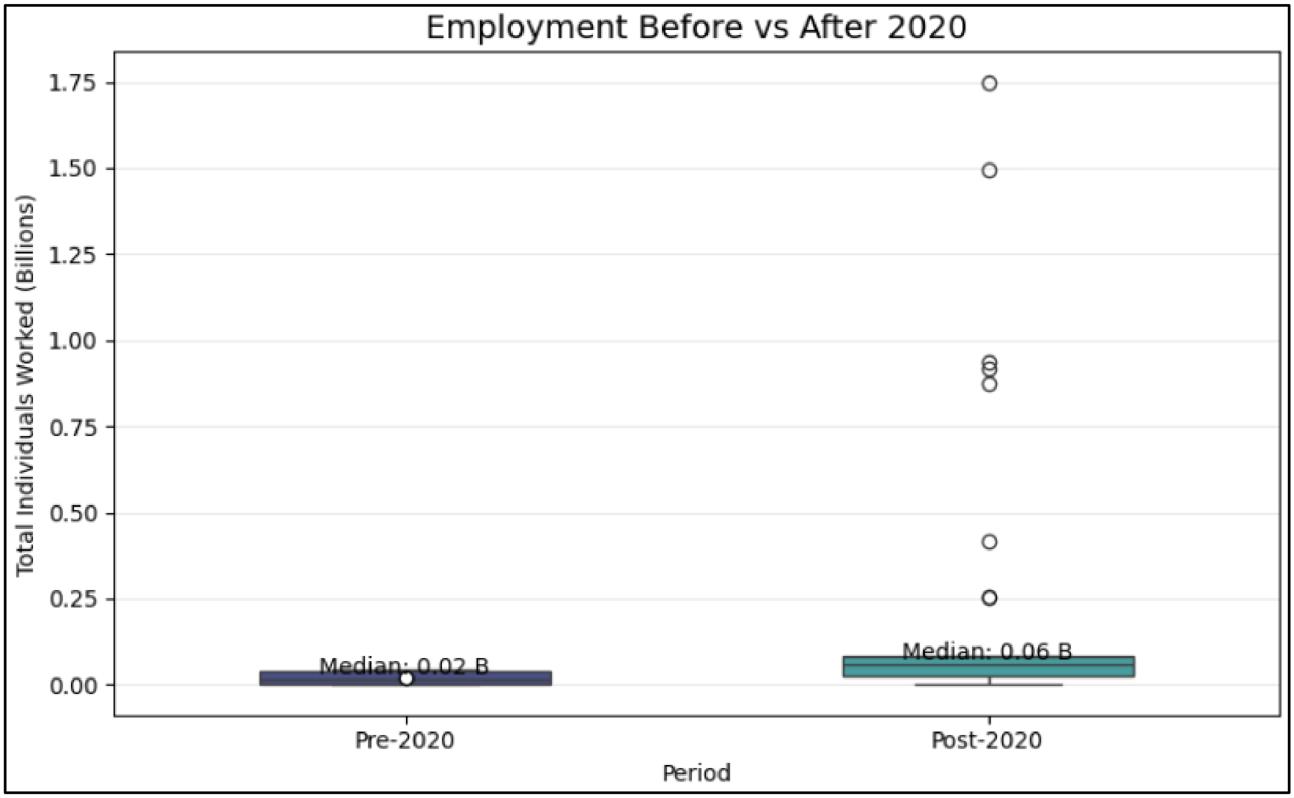


Expenditure is highest in East and South regions, indicating higher scale of operations. Northeast shows the lowest financial allocation.

Insights:

- East and South regions receive the highest average expenditure (~₹3 Cr**)**, indicating larger scale implementation of MGNREGA.
  This reflects stronger funding allocation and higher employment generation capacity in these regions.
- Northeast region shows the lowest expenditure (~₹0.4 Cr), significantly lagging behind other regions.
  This suggests limited financial allocation and lower scheme implementation.
- There is a wide gap between highest and lowest regions (~₹2.5 Cr difference), highlighting strong regional imbalance.
  This indicates unequal distribution of resources across the country.
- Overall, the data reveals clear regional disparity in funding and execution of the scheme.

13. Policy Impact Pre/Post 2020



Employment levels increased significantly after 2020, with higher median and variability. This reflects increased reliance on MGNREGA post-2020.

Insights:

- Median employment increases significantly from ~0.02B to ~0.06B, indicating nearly a 3× rise after 2020.
  This reflects a substantial expansion in employment generation under MGNREGA.
- Greater number of outliers in the post-2020 period shows higher variability in employment levels across states/years.
  This suggests uneven but overall increased demand for work.
- Wider distribution in the post-2020 boxplot indicates that more regions experienced higher employment levels.
  This reflects broader participation rather than isolated growth.
- The observed shift aligns with increased dependence on MGNREGA after economic disruptions like COVID-19.
  It highlights the scheme's role as a crucial safety net during crises.

12. Hypothesis Testing

1. Independent Sample t-Test (Pre-2020 vs Post-2020 Employment)

An Independent Sample t-test was performed to examine whether employment generation under MGNREGA differed significantly before and after 2020. The test compared the mean values of Total Individuals Worked for the two time periods. This analysis was conducted to validate the visual trend observed in the boxplot analysis. The obtained p-value was compared against the significance level of 0.05. The results indicated a statistically significant difference in employment generation between the pre-2020 and post-2020 periods, confirming the increased reliance on MGNREGA after 2020.

2. Pearson Correlation Test (Expenditure vs Employment)

A Pearson Correlation Test was conducted to analyse the relationship between total expenditure and employment generation under MGNREGA. The test measured the strength and direction of association between Total Expenditure and Total Individuals Worked. This was performed to statistically support the strong positive relationship observed in the correlation heatmap and scatter plot analysis. The correlation coefficient showed a high positive value with a statistically significant p-value (< 0.05). The results confirmed that higher expenditure is strongly associated with higher employment generation.

3. One-Way ANOVA Test (Women Participation Across States)

A One-Way ANOVA test was performed to determine whether women participation ratios differed significantly across states. The analysis compared the mean values of the Women Participation Ratio among multiple states included in the study. This test was conducted to validate the variation observed in the state-wise women participation visualization. The p-value obtained from the ANOVA test was evaluated at a 5% significance level. The results indicated significant differences in women participation across states, highlighting regional variation in gender inclusiveness under MGNREGA.

13. Expected Outcomes and Significance of the Study

The study aims to identify:

- Regions with consistent employment growth
- Regions heavily dependent on MGNREGA
- Budget allocation efficiency patterns
- Wage growth disparities
- Inclusion effectiveness
- Impact of economic shifts on rural employment

This project contributes to understanding:

- Regional inequalities in rural employment
- Policy responsiveness during crises
- Administrative efficiency patterns
- Relationship between budget allocation and employment outcomes.
- It supports evidence-based evaluation of rural development programs.

14. Conclusion

This study conducted a comprehensive analysis of MGNREGA using data preprocessing, feature engineering, exploratory data analysis, visualization, and hypothesis testing to evaluate employment generation, financial allocation, inclusiveness, and implementation efficiency across states and regions.

The analysis revealed a strong positive relationship between expenditure and employment generation, indicating that higher financial allocation contributes directly to increased rural employment. Wage expenditure formed a major share of total spending, highlighting the labour-intensive nature of the scheme. Significant regional and state-level variations were observed, with Andhra Pradesh and Rajasthan emerging as strong performers, while some regions such as the Northeast showed comparatively lower scale and participation.

Inclusiveness indicators demonstrated considerable participation of women and SC/ST communities, reflecting the scheme's role in promoting social equity. However, efficiency indicators showed that higher expenditure does not always ensure better utilization or work completion, emphasizing the need to evaluate both scale and efficiency separately.

Hypothesis testing statistically validated the key findings, confirming significant differences in employment after 2020, a strong relationship between expenditure and employment, and variation in women participation across states. Overall, the study highlights the important role of MGNREGA in rural livelihood generation while also identifying regional disparities and areas for improved implementation and resource utilization.